



Exploring blockchain technology as a governance layer for responsible artificial intelligence

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Received: 1 February 2026 / Accepted: 20 May 2026 / Published online: 1 June 2026
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Abstract

The rapid adoption of artificial intelligence (AI) across sectors has intensified the need for governance mechanisms that ensure its ethical, transparent, and accountable use. While governments, industry bodies, and civil society organizations have introduced numerous principles-based and values-driven frameworks, translating these high-level guidelines into practical governance practices remain a persistent challenge. This principles-to-practice gap is particularly pronounced given the increasing complexity, opacity, and adaptive nature of contemporary AI systems. This paper adopts a normative and conceptual approach to examine how this gap can be addressed. We identify the main risks associated with AI systems throughout their lifecycle and review existing governance initiatives to critically analyze the potential of blockchain technology as an enabling governance infrastructure in AI systems providing greater transparency, auditability, and accountability in AI development and use. We propose a conceptual governance framework that provides initial guidance on the types of governance-relevant data that can be stored on a blockchain layer of AI systems throughout their lifecycle. The paper offers a structured conceptual foundation for embedding governance mechanisms into AI systems, highlighting key technical, organizational, and societal considerations and contributing to ongoing debates on operationalizing AI governance. The proposed framework informs future empirical and design-oriented research. The paper has significant implications for advancing AI governance considering the merging of two transformative technologies and promoting the ethical and responsible use of AI.

Keywords Blockchain technology · Artificial intelligence · Governance · Data · Risks · Technology

1 Introduction

Artificial intelligence (AI) has rapidly emerged as a transformative force across industries, governments, public entities and wider societies, bringing unprecedented advancements in driving innovation, efficiency and analytical capabilities, and reshaping decision-making. Despite its tremendous potential, there are growing concerns about the accountability, transparency and ethical use of AI, urging governments, research institutes, industry bodies, and non-governmental organizations (NGOs) to develop appropriate AI governance mechanisms [1]. However, their efforts have been limited to promoting certain values, ethics and principles-based

frameworks, lacking guidance on how organizations can operationalize them. Several prominent research studies have examined the persistent “principles-to-practice” gap in AI governance [2–5], articulating common principles such as fairness, transparency, and accountability, and acknowledging the limited guidance available for operationalizing these principles in practical, organizational, or technical settings. Bridging this gap remains one of the central challenges of contemporary AI governance.

The development of AI governance models and mechanisms aimed at promoting AI innovative and responsible use, while addressing its potential risks and ethical and societal implications. So far, AI governance efforts have produced a fragmented and largely conceptual landscape of guidelines that often lag behind the rapid adoption of AI in practice. Failure in AI governance in sensitive domains, including healthcare, manufacturing, finance, and public services can have serious economic, social, and reputational consequences. Moreover, the adoption and interpretation

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of AI governance measures are deeply influenced by local socio-cultural, regulatory, and institutional contexts [6–9], underscoring the importance of considering these factors when developing AI governance approaches and solutions.

AI governance is less studied in the literature, with limited research investigating the development of practical AI governance mechanisms [3, 10–12]. One promising but underexplored avenue for operationalizing AI governance is through the use of blockchain technology. Originally emerged to solve the double-spending issue in financial transactions [13, 14], blockchain technology's distinctive characteristics has made the technology attractive to support AI systems governance across their lifecycle [15]. Few studies have explored blockchain's potential for specific AI systems governance, including its use to govern AI-generated content on digital platforms [16] and healthcare systems [17], emphasising the need for more research analyzing how blockchain can serve as an enabling infrastructure to operationalize AI governance principles and support enforceable accountability, transparency, and auditability. We aim to address this gap in this paper.

1.1 Theoretical positioning of the study

This paper sheds the light on the pressing gap between abstract principles of AI governance and the absence of practical enforceable mechanisms, acknowledging the importance of primarily developing conceptual foundations to support future technical development. It adopts a normative and conceptual approach [18], to investigate how AI governance systems should be designed based on shared ethical principles and desired outcomes in order to develop system-wide governance infrastructure for AI. We draw on insights from governance and socio-technical systems literature [3, 8, 10, 11]. Rather than developing a formal theory or technical artefact, the paper proposes a conceptual governance-oriented framework aimed at supporting the operationalization of AI governance principles in practice.

By introducing a governance infrastructure perspective, the paper originally conceptualizes blockchain as a governance layer for AI systems that supports inter-and intra-organizational traceability, coordination, and accountability across the AI lifecycle, rather than solely being used as a tool for audit, compliance, or data management. In this role, blockchain has the potential to enable continuous, lifecycle-wide governance, supporting not only ex-post auditing but also ongoing monitoring, validation, and accountability. By mapping AI system risks and governance requirements to the core characteristics of blockchain technology, the proposed framework advances existing literature by offering a structured pathway for translating abstract governance principles into more actionable and system-embedded practices.

This study makes three contributions to AI and governance literature. First, it extends the underdeveloped AI governance literature by analyzing the risks of AI systems at each stage of their lifecycle and identifying how blockchain can advance practical governance at each stage. Second, it provides conceptual guidance for organizations seeking operational mechanisms to govern AI systems throughout their lifecycle, positioning blockchain as a potentially adaptable governance infrastructure across different organizational sizes, sectors, and contexts. Third, the paper highlights the type of data that can be stored on the blockchain governance layer of AI systems, which is foundational for future empirical and technical research. The paper does not claim blockchain is a panacea for AI governance and highlights key considerations for the effective use of the technology, seeking to stimulate critical reflection and interdisciplinary dialogue on how emerging technologies can bridge the principles-to-practice gap. These insights are valuable for researchers, practitioners, and policymakers aiming to promote ethical and responsible AI deployment.

The paper is organized as follows: we first examine the risks associated with AI systems throughout their lifecycle (Sect. 2) and review existing AI governance initiatives and their limitations (Sect. 3). We then analyze the core capabilities of blockchain technology and its potential as a governance technology for AI systems, including a critical analysis of three case studies of real blockchain-enabled AI governance platforms (Sect. 4). Insights derived from this staged analysis form the basis for the development of a conceptual framework that positions blockchain as a governance infrastructure for AI, including guidance on the types of data that can be recorded and validated on the blockchain layer (Sect. 5). Key considerations for blockchain use in AI governance are highlighted in Sect. 6. Finally, we discuss implementation challenges, societal and contextual considerations (Sect. 7), and future research opportunities (Sect. 8).

2 The main risks associated with AI systems throughout their lifecycle

An AI system is “a machine-based system that, for explicit or implicit objectives, infers, from the input it receives, how to generate outputs such as predictions, content, recommendations, or decisions that can influence physical or virtual environments” [19]. The AI system lifecycle consists of four stages, including (1) design, data and model, (2) verification and validation, (3) deployment, and (4) operations and monitoring [19]. The different stages of the AI system lifecycle are iterative, stressing the importance of having an

audit trail to track the development and use of AI systems and their data [10].

AI systems can be deductive or generative. Deductive AI systems analyze large datasets to identify patterns and derive conclusions and decisions, while generative AI systems learn from existing data to generate new content [20]. The risks of AI systems can be related to three main elements: input data, AI model parameters and design, and system output. We briefly discuss the main risks associated with each element below.

2.1 AI system input data risks

AI systems can analyze vast amounts of data, learn from it and identify patterns, relations, predictions and decisions based on the input data. When constructing the AI system model, an initial dataset called training data is used to develop the model parameters that meet the system's objectives [10]. The training data is a crucial element in AI systems development that helps ensure the model will perform as intended in its end application environment and minimizes any risks of adverse outcomes [21].

Some of the risks associated with the training data include its limited representation of all possible scenarios that an AI system might be subjected to in its real application environment. The AI system might be subjected to unanticipated situations or uncertainties that can lead to unexpected system behaviour [10]. This is particularly risky in safety-critical AI applications in healthcare, autonomous vehicles and defence. Furthermore, any biased correlations evident in the training data can lead to the creation of algorithmic bias in the model [20] and biased system outputs [6, 8, 21]. Bias in any of the AI system elements might lead to detrimental, discriminatory and/or unethical outcomes. Training data screening and the removal of any sensitive data can help reduce this risk [10]. The use of simulation can also help expose the system to a large number of different scenarios to minimize unexpected AI system behavior risks. The use of synthetic training data can further help address the limitation of finding sufficient training data for the model to avoid bias [20]. However, there is a need to manage access to the training data and to monitor and validate its use for developing appropriate AI models.

2.2 AI model parameters and design risks

AI systems continuously learn from the data to improve the model's precision and outcomes. The system can analyze data from different sources, including Internet of Things (IoT) and other information technology (IT) systems. It can process big data, personal data and social media data requiring more attention to data governance. The large amount

of data the system is subjected to can lead to changes in the programmed rules overseen by the system users or to an algorithmic bias which can have significant implications [20]. Bias in AI systems can also be intentional by designing the AI model parameters in a way that includes or excludes certain outcomes. These can be evident in AI use in employment processes such as applicants' selection processes, analysis of criminal activities, and personalization processes which raises questions about the ethical use of AI in such applications [21]. This is in addition to the inherent complexity and opacity of advanced AI models, creating a black box and limiting their explainability and the ability to understand the system behaviour and how it changes over time [10, 21]. All of these can lead to unintended outcomes, hazardous to its users. Furthermore, the loss of human control with the increased autonomy of AI models can have several significant implications on the accountability and liability of decisions made by AI systems [21, 22].

2.3 AI system outputs risks

Most of the associated risks with AI systems relate to the system output and how it is used or informs subsequent decisions. Overreliance on AI for decision-making has significant risks as AI outputs can be influenced by the quality of the training data used to design the model, the input data and the model parameters. The recent advancement of AI technologies and their increased autonomy have made it difficult to predict and control the model evolution and performance, causing serious concerns related to AI governance and its responsible and ethical use. Furthermore, despite the potential of AI in transforming society, the implications of AI systems' risks require immediate attention as they can have serious economic and social impacts. Economically, the use of AI systems is transforming the job market with increased automation, and the differences between the countries' job markets in terms of skills requirements and availability which will also magnify depending on their stage of AI system adoption and use [23, 24]. These can also lead to societal implications by increasing the unemployment rate, damaging physical and mental health, infringing on individuals' privacy and undermining human rights [25]. Further risks include the weaponization and malicious uses of AI that can have detrimental consequences for societies [6, 8]. All of these stressed the need for developing new approaches for AI governance that encourage responsible use of AI and minimize associated risks.

3 Major developments in the governance of AI

3.1 Overview of major AI development priority areas

AI is not developing along a single global trajectory as AI applications and governance approaches vary across industries and countries. Some sectors have witnessed advanced applications of AI such as the medical sector in which several successful AI applications in analyzing patient data, disease diagnosis, predictive medicine, clinical decision support systems [26], and robotics [9, 27] are evident. AI has also played a crucial role in the development of autonomous vehicles [28, 29], intelligent wind turbines [30], aerospace predictive maintenance systems [31], textile material testing [32], and farming and irrigation systems [7] to name a few. However, these developments are not equally evident across the different regions as different countries have different priorities for AI depending on their societal needs, economic priorities and cultural value. For instance, public transport and healthcare are top priorities in Japan and Singapore, supporting the development of AI in surgical robots and autonomous vehicles to tackle labour shortages and ageing populations [8]. Nations with strong industrial manufacturing sectors, including Germany, have focused on integrating AI to enhance industrial automation and productivity [6]. The governments of China, the US and Russia have made significant investments in AI use to develop their military assets and strengthen their national defence [8], while the government of Spain secured a large investment with Microsoft to foster the deployment of responsible AI in public administration and strengthen the country's national cybersecurity resilience [33]. These differing priorities illustrate that AI systems emerge within specific political, economic, social and cultural conditions rather than being universally adopted in the same way. Recognizing these variances in AI contextual factors is crucial for addressing its associated risks as they could vary significantly depending on the application domain and scope, local norms, institutional trust and regulatory requirements. For example, AI in healthcare may be seen as augmenting medical expertise in some countries with advanced resources compared to others and can be strongly influenced by contextual factors such as patients' trust in AI [34]. In agricultural, AI priorities may vary between communities considering their technical expertise, technology affordability, investment requirements, and/or need for sustainability compliance [35]. These variations highlight that "one-size-fits-all" governance approaches for AI are unlikely to succeed, stressing the need for more adaptable solutions to ensure effective oversight and responsible use of AI.

3.2 Overview of current AI governance initiatives

The diversity of AI application areas and level of adoption poses significant challenges to the development of unified approaches for AI governance. Governments and leading organizations have been working on various initiatives [36]. For instance, Canada is one of the Digital Nations actively working on AI governance and developed guidance for federal institutions on the use of GenAI and responsible AI in government [37]. The European Union (EU) developed the first AI regulation indicating that AI systems are regulated based on the level of risks of its specific application [38]. The EU AI Governance Act has been developed to ensure the safe use of AI in Europe, protecting fundamental human rights and democracy and supporting businesses' use of AI [38]. The UK released its pro-innovation approach to AI regulation policy white paper in March 2023 guided by five principles for responsible development and use of AI in all sectors, including: (1) safety, security and robustness, (2) appropriate transparency and explainability, (3) Fairness, (4) Accountability and governance, and (5) contestability and redress [25]. The developed framework takes a sector-led approach, delegating the development of AI governance mechanisms to existing regulators of the specific sector [39]. China has initiated a call for a global AI governance inviting all countries to exchange information and co-operate on the governance of AI [40]. The US released a blueprint for an AI bill of rights, including a set of five principles and associated practices: (1) safe and effective systems, (2) algorithmic discrimination protection, (3) data privacy, (4) notice and explanation, and (5) human alternative, consideration and fallback [41]. In 2019, the OECD developed key principles for AI use that promote innovative and trustworthy use of AI and respect human rights and democratic values [19]. The World Economic Forum (WEF) has launched the AI Governance Alliance to promote the transparent and responsible design and use of inclusive AI systems [42]. More recently, the United Nations launched two bodies for AI governance including the Independent International Scientific Panel on AI and the Global Dialogue on AI Governance [43].

At an organizational level, technology providers, industry and research institutes have also been actively working on AI governance, developing mechanisms and frameworks to support the ethical and responsible use of AI. Technology companies like IBM developed an AI governance framework that was adopted by leading companies such as Mahindra Tech and Deloitte [44]. Google AI developed its own set of principles to demonstrate its commitment to responsible AI [45]. However, AI corporate governance is not easy, raising many challenges for large corporations such as OpenAI [46, 47]. These initiatives demonstrate significant momentum

toward establishing norms, expectations, and shared values for AI use across sectors and societies.

3.3 Limitations of current AI governance approaches

AI governance approaches to date can be classified into four categories, including (1) promotion of relevant ethics, norms and values frameworks, (2) deploying legislative measures and forming regulatory bodies and jurisdictions to govern AI technology, (3) raising awareness and investigating the implications and possible solutions to AI use, and (4) developing technical solutions to address certain issues raised by AI technology [6]. These four areas are not equally developed, raising the need for further research and more comprehensive approaches to AI governance.

Despite the growing development of AI governance frameworks, many remain difficult to translate into practice. High-level principles such as transparency, fairness, and accountability are frequently articulated but rarely accompanied by mechanisms for operational implementation [2]. For instance, audits of large-scale AI models have revealed that model training datasets are often undocumented or only partially traceable, making it difficult to verify data provenance, bias mitigation, or compliance with consent and copyright requirements [48]. Recent copyright lawsuits against generative AI developers further illustrate how unauthorized, unclear or missing dataset lineage complicates regulatory enforcement and ethical accountability [49]. Similarly, although regulatory instruments such as the EU AI Act outline oversight expectations, they provide limited guidance for how organizations should continuously monitor models as they evolve, drift, or are repurposed over time [38]. These gaps highlight the need for defining clear technical requirements to embed governance principles and compliance requirements into the design of AI systems [50]. There is a need for adaptable technical solutions capable of recording, tracing, and validating system operations in a secure and tamper-proof manner to ensure compliance with evolving regulations.

AI governance has been a prominent topic in the academic literature but with limited yet growing research investigating the development of AI governance mechanisms. Mäntymäki et al. (2023) proposed an hourglass model of organizational AI governance including three layers namely environmental, organizational and AI systems. Schmit et al. (2023) proposed a governance model that combines copyleft licensing and a “patent troll” model to manage AI intellectual property (IP) rights and ensure ethical governance of AI. Janssen et al. (2020) proposed a framework for data governance of AI systems that takes into consideration both the system algorithms and data. Other studies aimed at studying

and mapping existing governance initiatives [6, 39, 51–54]. However, the developments of AI governance mechanisms lag behind its fast-paced implementation in practice, raising the need for more research developing practical governance solutions that target the core areas of AI systems risks and enforce compliance with evolving guidelines.

4 Blockchain core capabilities and relevance to AI governance

4.1 Overview of blockchain core capabilities

Blockchain is a distributed ledger technology, in which transactions are stored in blocks and each block is time-stamped, cryptographically identified and links to the block that precedes it in the system [14]. Consensus mechanisms are used to validate the blocks before adding them to the ledger and ensure that all nodes have access to the same ledger status [55]. Smart contracts, which are programmable codes that can execute transactions autonomously if predetermined conditions are met, bring additional capabilities to blockchain and support a wide range of blockchain applications across different sectors [56].

The blockchain technology is a combination of technologies that collectively bring unique capabilities. The decentralized infrastructure of blockchain’s distributed ledgers helps tackle the issues of centralized information sharing and being subject to a single point of failure. Digital timestamping and the use of cryptography support the immutability of blockchain data and help tackle issues related to data manipulation, falsification and security concerns, particularly when sharing data among different nodes or participants [14]. The chronological record of blockchain transactions can help in the traceability and verification of blockchain transactions and their history, holding the network participants accountable for their data. This is particularly important for dispute management, and auditing purposes. The use of consensus mechanisms ensures agreement on the ledger status and on the governance mechanisms used to validate and add blocks in the network. Smart contracts can help design customized rules that support blockchain data access and control [57]. The use of consensus mechanisms and smart contracts differ depending on the blockchain purpose and application area, and they are key aspects of blockchain governance.

4.2 Blockchain potential for AI governance

The core characteristics of blockchain are what make the technology attractive for AI governance. Most existing AI governance initiatives and principles aim at improving the transparency, understandability and accountability of AI

systems. This requires the availability of an audit trail of how the AI system has been developed and used throughout its lifecycle [58]. Blockchain's ability to create transparent, traceable, and immutable records of transactions can help address many stressing issues related to AI governance [15]. The technology can support the creation of an audit trail of AI systems throughout their lifecycle, ensuring the transparent, ethical and responsible development and use of AI [59].

Compared with alternative AI governance approaches such as federated learning (FL), trusted execution environments (TEEs), and traditional audit logging systems, blockchain offers distinct governance capabilities that extend beyond privacy preservation or secure computation. FL techniques have been used to support AI governance in several sensitive contexts like healthcare [60] as they enable decentralized model training without sharing raw data [61]. While these techniques help in reducing costs, maintaining privacy and enabling decentralized collaboration [61], they have major limitations in terms of lacking built-in mechanisms for immutable auditability of the data used, cross-organizational accountability, and governance transparency [60]. Similarly, TEEs provide secure and confidential execution environments for AI computations that rely on trusted and isolated hardware infrastructures [62]. While TEEs can guarantee data security and model integrity, they offer limited visibility into broader governance processes and require large investments on hardware. Traditional audit logging systems support internal monitoring and compliance but are typically centrally controlled and therefore remain vulnerable to alteration, selective disclosure, or disputes over data integrity [63]. Blockchain complements these approaches by enabling the creation of a shared, tamper-resistant, and verifiable audit trail across multiple stakeholders throughout the AI lifecycle. Through decentralized recordkeeping, cryptographic verification, and programmable smart contracts, blockchain can support continuous monitoring, provenance tracking, compliance validation, and coordination across organizational boundaries. Similarly, blockchain can benefit from the capabilities of existing governance approaches such as with the use of FL techniques for monitoring model development and TEEs by offering secure off-chain data computations [62]. Therefore, blockchain should not be viewed as a universally superior solution but rather as a part of a broader socio-technical governance ecosystem that can complement existing governance and security technologies depending on the specific AI application and context.

Few studies in the literature have explored blockchain capabilities in governing AI systems. Taddese et al. (2025) investigated the issue of managing complex and vast datasets generated by advanced AI models in genomics and microscopy. Their review identified the potential of

blockchain technology in addressing bias, fairness and privacy concerns related to AI use, securing data access and control, protecting data, and enhancing data traceability, accountability, verifiability and ethical compliance [17]. Yang et al. (2024) propose methods for governing AI generated content on digital platforms using blockchain technology. Their work indicates the capabilities of blockchain decentralized architecture, cryptography, and traceability in supporting the governance of AIGC [16]. Their research developed several frameworks for blockchain-enabled content tracking and validation, owner identification and assurance, and content security and regulatory governance. The proposed frameworks were tested using real time data yielding favourable performance. Alruwaili (2020) proposed the use of blockchain technology and AI multi agent system to ensure the privacy and security of electronic health records [64]. Asif et al. (2023) explored how blockchain smart contracts can support the responsible use of AI considering various technical aspects of their use [57].

The recent developments in decentralized identity management further enhance blockchain's potential for AI governance. For example, blockchain-enabled self-sovereign identity (SSI) frameworks [65] and verifiable credentials [66], enable individuals, organizations, and even AI systems to authenticate their roles and responsibilities in a cryptographically secure and privacy-preserving manner. They have the potential to address a core weakness in existing AI governance frameworks by enabling the traceability of who contributed what, when, and under which conditions without exposing sensitive information. For example, datasets used to train AI models can be accompanied by verifiable credentials confirming consent provenance, licensing agreements, or ethical review approvals, while AI models and their versions may be issued credentials documenting audit outcomes, safety assessments, and deployment contexts. Integrating SSI and VCs into blockchain-based governance thus allows compliance, provenance, and accountability claims to be machine-verifiable, supporting operational implementation of ethical principles in dynamic and distributed AI ecosystems.

4.3 Real implementation of blockchain for AI governance

Although research on blockchain-enabled AI governance remains nascent, several technology firms have begun implementing blockchain infrastructures that operationalize traceability, auditability and accountability mechanisms for AI systems. The below cases illustrate the practical feasibility of integrating blockchain into AI governance workflows. Table 1 provide comparative analysis for the selected cases.

Table 1 Comparative analysis of emerging blockchain-enabled AI governance platforms, highlighting governance focus, core governance mechanisms and enabled outcomes, strengths, and limitations

Platform	Prove AI	EQTY Lab	Humans.ai
Governance Focus	Lifecycle monitoring and auditability of AI systems	Verifiable and compliant AI system governance across lifecycle and supply chains	Decentralized governance of AI agents and ecosystem-level coordination
Core Governance Mechanisms	Hybrid blockchain logging (Hedera); telemetry integration (OpenTelemetry, Prometheus); cloud-based key management (AWS KMS); metadata and prompt tracking	Blockchain-based registration (Hedera Consensus Service); cryptographic signing of AI lifecycle stages; developed solutions include Verifiable Compute; AI SBOM and AI Guardian tools	Blockchain-based AI orchestration (“blockchain of AIs”); agent competition and peer evaluation; token-based governance; (historically) PoH and AI NFTs
Governance Outcomes Enabled	Enables traceability of datasets, model evolution, and AI interactions; supports auditability, compliance, and IP protection through verifiable records	Enables tamper-proof audit trails of AI processes; supports regulatory compliance and supply chain governance; strengthens trust in model development and deployment processes	Enables decentralized participation in AI governance; supports transparency through lifecycle records; introduces performance-based accountability via agent evaluation mechanisms
Strengths	Strong integration with existing enterprise infrastructure; comprehensive lifecycle data capture; supports observability for generative AI systems	Strong focus on cryptographic verification and compliance; integration with hardware-level validation; alignment with regulatory requirements; supports multi-level governance (models, agents, supply chain)	Innovative ecosystem-level governance model; promotes decentralization and participatory governance; supports traceability and AI lifecycle visibility
Limitations / Challenges	Reliance on cloud infrastructure introduces partial centralization and potential vendor dependency; interoperability and integration complexity; limited evidence of effectiveness at scale; governance remains dependent on trusted intermediaries	High infrastructure costs due to reliance on specialized hardware (e.g., Intel, NVIDIA); potential performance latency from cryptographic layers; cannot address data quality issues (“garbage in, garbage out”); risk of infrastructure lock-in	High technical complexity limits accessibility; governance effectiveness depends on participant incentives; unclear current role of PoH and AI NFTs; evolving governance model

4.3.1 Prove AI

Prove AI is one of the most prominent platforms offering governance tools for AI systems. The company integrates various technologies to capture and provide comprehensive data about AI models throughout their lifecycle. The main objective of its solutions is to help organizations maintain crucial oversight over their AI models development, data usage and performance, thereby helping to mitigate associated risks. The company’s flagship AI governance product is built on Hedera’s decentralized blockchain network for securely logging and storing of various data types such as AI training datasets, models metadata and context information, generative-AI prompt sessions and other machine learning data [67]. Amazon Web Services (AWS) and key management service (KMS) are used to help generate key pairs for signing transactions on both private and public blockchain architectures. Prove AI also offers industry-standard observability tools for generative AI models based on OpenTelemetry and Prometheus open sources for telemetry data collection and storage capabilities (<https://proveai.co>

[m/faqs](#)). Prove AI also supports IP protection by enabling the traceability and verification of copyright information to their sources in case of IP infringement [67].

While Prove AI provides a structured and innovative approach to capturing and managing AI lifecycle audit data, whether on blockchain ledgers or alternative storage systems depending on organizational preferences, its governance model relies heavily on integration with multiple cloud infrastructures. This reliance may introduce security, integration and interoperability challenges, contribute to partial centralization, and limit the extent of decentralized accountability. Moreover, despite its comprehensive audit capabilities, there is limited publicly available evidence demonstrating its effectiveness in large-scale, real-world deployments. This raises important questions regarding scalability, interoperability, and the degree to which such platforms can operate independently of trusted intermediaries.

4.3.2 EQTY Lab

EQTY Lab is another real example of how organizations can use blockchain capabilities to govern and enhance the transparency of AI systems development and use. The company collaborates with leading organizations such as Nvidia, Intel and Hedera to develop customized and multi-level AI governance tools that enables organizations to track and govern their AI systems throughout their lifecycle [68]. EQTY Lab uses blockchain-based Hedera Consensus Service to enable the registration of every step in the AI system's lifecycle using cryptographic signatures, including initial datasets and data sources, training parameters, model modifications and deployment, creating an immutable, auditable and verifiable chronological record of the model lifecycle [68]. The company's innovative Verifiable Compute solution aims to establish truth at the core hardware level by cryptographically signing every stage of the AI lifecycle, providing tamper-proof auditable records of AI systems behaviour. Verifiable Compute is developed in collaboration with government agencies to enforce compliance with emerging and evolving AI regulations [68, 69]. EQTY AI offers a variety of governance solutions including AI software bill of materials (AI SBOM) that ensures AI models compliance with organizational policies and regulations and enables the governance of AI models, data and agents across the supply chain, and AI Guardian that is specifically designed to help govern AI agents, continuously monitoring their configurations and performance. These set the company apart as it does not only provide a practical solution for governing AI models but also helps organizations comply with and navigate the complex landscape of AI regulations.

EQTY Lab offers robust AI governance solutions that include significant hardware dependency on high-end Intel and NVIDIA chips, creating high entry costs and potential infrastructure lock-in. It raises important considerations regarding the feasibility of the cryptographic verification layer required for ensuring greater transparency and traceability of AI systems as having an added cryptographic layer to every AI computation may introduce performance latency, potentially impacting high-speed applications. Furthermore, although the platform offers various tools that support the verification of the integrity of the process, it cannot inherently eliminate "garbage in, garbage out" scenarios involving biased training data or model hallucinations.

4.3.3 Humans.ai

Humans.ai integrates AI and blockchain in an innovative way to develop powerful intelligent systems. The company adopts a decentralized approach by combining blockchain infrastructure with AI system intelligence and orchestration,

enabling the creation, evaluation and evolution of multiple AI agents within a shared secured ecosystem. Its current model emphasizes competition, peer evaluation, and continuous improvement among AI "agents," where outputs are assessed and ranked to determine performance and accountability [70]. The company's developed decentralized blockchain-AI platform, named "blockchain of AIs", supports the development, operations and governance of AI models, utilizing blockchain capabilities to address the transparency, auditability and ethical use challenges of AI systems [71]. The use of blockchain supports the creation of a secure tamper-proof digital records of AI systems' entire lifecycle, from where its data came from to how it was trained and used.

Humans.ai has been associated with governance mechanisms such as Proof-of-Human (PoH) consensus and the use of AI-enabled NFTs, both aimed at enhancing accountability, ownership, and traceability in AI systems. Earlier descriptions of the platform emphasise PoH as a mechanism supported by biometric validation and human oversight, alongside the use of AI NFTs, where AI model data are encapsulated into tokenized assets to enable governance, provenance, and access control [72]. However, recent platform positioning places less explicit emphasis on these features, suggesting a potential shift toward broader decentralized and token-based governance models centred on AI agent orchestration and performance-based evaluation. As a result, the current operational role of biometric-based validation and NFT-enabled governance for AI systems remains unclear, reflecting the evolving nature of governance mechanisms in emerging blockchain-enabled AI ecosystems.

5 Blockchain technology as a governance infrastructure for AI

Despite the growing interest in blockchain potential for AI governance in research and industry, this area is significantly underdeveloped, requiring more exploration of how the technology can be used to support AI governance in various applications. Blockchain technology has great potential in providing a unified approach for AI governance that can be used regardless of the organizational size, geographical remit or context and scope of the application. Organizations can use blockchain internally to govern their own development and use of AI in their business operations and customer services, and/or externally with their customers and technology providers to ensure the proper deployment and operations of their AI systems, or with auditing firms, regulatory or certification bodies to verify and validate their compliance with relevant laws and regulations.

Based on the previous analysis, in this section, we propose a governance-oriented conceptual framework highlighting the role that blockchain can play as a governance layer in AI systems by considering how the technology can address the main risks associated with AI systems input data, model and parameters and outputs, and the type of data that can be stored on the blockchain at each stage of the AI system lifecycle (Fig. 1).

(1) AI system design, data and model development

AI governance at this stage is very critical for the transparent and ethical development of AI systems. Depending on the purpose of the AI system, training data is used to inform the development of the model algorithms and parameters and to produce relevant outcomes. Some of the key risks at this stage are having unlicensed, biased or discrimination parameters in the training data or the model algorithms that could influence the model outcomes and decision-making. The lack of transparency about the AI system development stage hinders the

explainability of AI systems which might compromise users' rights. It can also affect the authenticity of AI outcomes and raise IP challenges [73], and the accountability and verifiability of AI systems in case the system produces a biased outcome or performs in an unintended manner [21]. Blockchain technology can be used at this stage to enable AI governance by providing a mechanism to (1) record and verify the training data used for AI model development, ensuring the removal of sensitive data, biases, discriminations or intentional patterns in the data, (2) maintain a tracked record of changes in the training data which can help better understanding of how the model outcomes change with changing input data, (3) record and track changes in the model parameters and used algorithms to ensure they are consistent with the intended purpose, (4) improve accountability in AI systems by providing an audit trail of the development cycle of AI systems including information about data providers, training data sources and authorization, the model algorithms used and parameters, and the intended use and outcomes. The use of blockchain

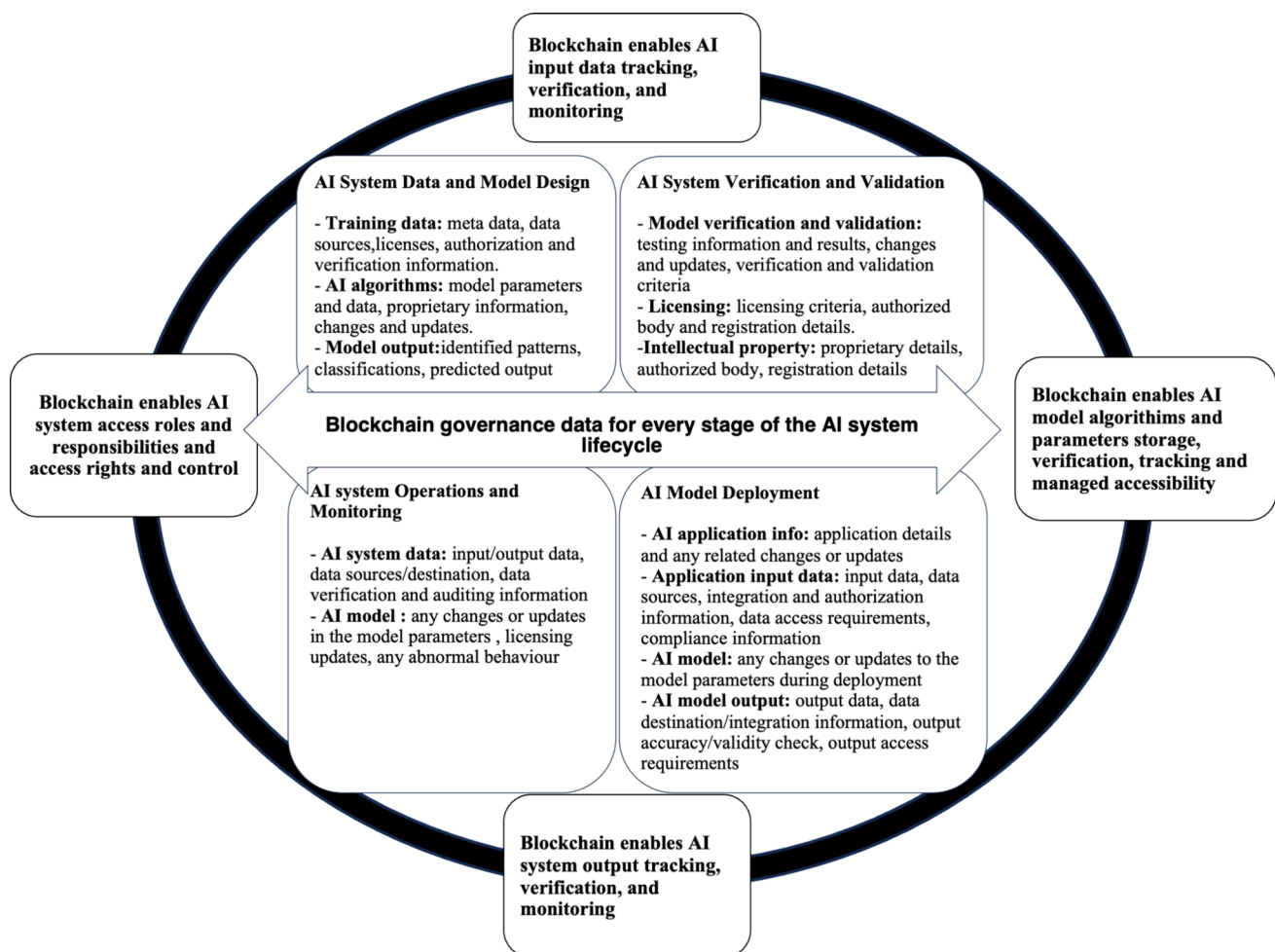


Fig. 1 Blockchain-enabled AI governance throughout the entire AI system lifecycle

can help ensure the transparency of AI systems development by enhancing data tracking, security, validity, and verifiability and addressing risks associated with the training data or the model algorithms. Given that the size of the data that the model uses can be massive and varies depending on the application, only a hash of the data or meta-data can be stored on the blockchain [59].

(2) AI system verification and validation

By creating a secure and traceable audit trail for AI systems development, blockchain technology can support the governance of AI systems during the verification and validation stage in several ways: First, the audit trail ensures that AI systems are developed in a transparent and ethical way to achieve their intended purpose. Second, in the case of any unintended outcomes, the audit trail can be used to verify and validate the AI system data and outcomes since its development and identify any changes in the model outcomes or discrepancies in the model or its data. Third, the use of blockchain also supports the governance of people's interaction with the AI system. Only authorized parties can have access to the model data and algorithm parameters which can enhance data privacy and security and improve accountability in AI systems [59]. Fourth, the audit trail can be checked by a third party such as a technology developer, an auditing firm, or a licensing or regulatory body to verify and validate the AI system development and authorize its use for its intended application. Once the AI system is verified, the model parameters, data characteristics and predicted outcomes can be recorded on the blockchain along with the system authorization and obtained license. Blockchain technology can be used to manage and track AI systems authorization, licensing, and IP by providing a secure and tamperproof platform for managing and verifying interactions with AI systems. Blockchain use for IP tracking and management has been evident in different contexts [74]. Developing rigorous mechanisms for AI IP governance is crucial to promoting the innovative development of AI applications while protecting the rights of developers, creators and users [73].

(3) AI system deployment

The deployment stage is one of the most challenging stages of the AI systems lifecycle, in which the AI system is integrated and put into use in its intended application area and is fed with the application data to produce practical decisions. AI governance is required at this stage to ensure the proper deployment of the AI system

in compliance with contractual, organizational, regulatory and societal policies. AI system integration might require changes to the ML algorithm to support data flows between the different IT systems. These changes need to be verified and validated. The use of blockchain at this stage can ensure that the deployed model is authorized, licensed and verified for its intended application and any adjustments made are verified and recorded on the blockchain to monitor the system performance. Blockchain records can be used during the testing phase to assess how the model will perform with real data compared to the training data used to develop the model and to ensure that the outcomes are acceptable, valid and verified.

(4) AI system operations and monitoring

AI systems vary in their level of autonomy and adaptiveness after deployment [19]. With the continuous use of AI systems, particularly deep learning and large language model (LLM) capabilities, the AI system can adapt to the changes in the input data. This raises several risks related to having unforeseen outcomes that might misinform decision-making, leading to unintended consequences, and even be harmful to end users. Therefore, governance of AI systems extends beyond the deployment stage to the entire use life of the AI system. The use of blockchain and smart contracts can help in governing the deployment, operations and monitoring of AI systems [57]. Blockchain can be used to regularly store the decisions and outcomes of AI systems since its deployment, providing an audit trail that can be checked to ensure that the system is performing as intended and monitor any changes to the AI model. These regular checks can be conducted to a sample of the input data, model algorithms and/or resulting outcomes depending on the specific application and degree of associated risks. The use of smart contracts can help prevent extreme or inaccurate outcomes from being acted upon, by continuously reviewing the AI system decisions and outcomes and ensuring it is within a normal and acceptable range. Any abnormalities in the system outcome can be identified and alerted by the smart contracts, requiring human intervention and an audit.

Table 2 summarizes how blockchain can help address AI systems risks throughout their entire lifecycle. We also provide a brief illustrative example of how blockchain can help govern AI diagnostic system in healthcare in Sect. 5.1.

Table 2 Blockchain potential for AI systems governance

AI systems life-cycle stage	AI governance element	Associated risks	Potential of blockchain technology
AI Design, Data and Model Development	AI input data, training data governance	Training data includes discrimination or sensitive data Use of unlicensed or unauthorized data	Provide a secure and traceable record of the training data used to develop the AI model Any changes in the training data can be tracked and verified Help ensure no sensitive, biased or discriminatory data is evident in the training data
	AI algorithms development and tracking	Blackbox problem and inability to understand the behaviour of the AI system AI model complexity and explainability Having the model designed in a way that leads to biased outcomes Errors in developed algorithms	Provide a secure and traceable record of developed AI algorithms and model parameters AI model parameters can be regularly checked and stored in the blockchain to ensure accuracy, and consistency and track any changes in the model The existence of a trail record for the model programming rules and parameters can help in understanding and tracking the behaviour of the system over time and for different types of input data
	AI model outcomes	Biased outcomes Unpredicted outcomes can be harmful	The range of possible and acceptable outcomes can be identified and used to design smart contracts that prevent any extreme and abnormal outcomes A traceable record of model changes can help prevent and mitigate adverse outcomes
	AI system verification and validation	Training data is invalid and does not meet the purpose of the application AI model does not yield targeted outcomes	Having a traceable record of the training data and model parameters can be used to verify their validity for the AI system development and targeted application
AI Verification and Validation	AI licensing	Unauthorized use of data and AI systems Misuse of AI in a harmful way	The blockchain audit trail developed during the design and development stage is used to authorize and license the final model License details are stored on the blockchain to ensure authorized and tracked use of AI systems
	AI IP protection	Misuse or unauthorized use of AI IP copyright infringement	Blockchain distributed ledger can be used to authenticate, manage, track and monitor intellectual properties of AI systems
AI Model Deployment	AI model application	Inappropriate application of AI system	Having the data about the AI system in a secure blockchain platform can be used to verify the system parameters and its suitability for the targeted application
AI Operations and Monitoring	Input data	Changes in the input data that can lead to unintended outcomes	Blockchain can keep track of the input data to provide a traceable record to investigate in case the model has unexpected behaviour
	Access to algorithms	Unauthorized/intentional manipulation of the AI model	Blockchain data access rules can help ensure only authorized parties have access to the AI system algorithms Blockchain can help provide secure and authorized access to AI algorithms that can be used to track any changes in the deployed model over time
	Use of AI System	Unauthorized use of the AI system	Blockchain's audit trail provides a reference to the AI system through its lifecycle

5.1 Illustrative use case: governing AI diagnostic systems with blockchain

The three case studies discussed in Sect. 4.3 demonstrated the technical feasibility of developing blockchain infrastructure for AI governance. Here, we briefly illustrate a practical example to show the value of blockchain-enabled AI governance in AI diagnostic systems in healthcare. These systems increasingly support clinical decision-making by analyzing medical images, patient histories, and genomic data [17]. However, concerns frequently arise regarding data provenance, patient consent, model bias, and accountability when AI-generated recommendations influence medical outcomes [34]. In such contexts, blockchain can provide a verifiable and tamper-resistant audit trail that records how patient data was sourced (AI input data), when and how the AI model was trained or updated (AI model data), and which actors accessed or modified the system at different points in the workflow (AI system access) [17]. This can help resolve core ethical and social dilemmas related to trust, transparency, and responsibility by enabling clinicians, patients, and regulators to trace decisions back to their origin, verify compliance with consent and privacy requirements, and identify accountability in the event of harmful outcomes. Moreover, incorporating SSI techniques and verifiable credentials can allow patients to maintain control over their own health data while still enabling secure, authorized access for legitimate medical use. In this example, blockchain does not replace existing governance mechanisms but strengthens them by making AI decision processes more visible, auditable, and aligned with patient rights and broader societal expectations for ethics in healthcare.

6 Key Considerations for blockchain use in AI governance

The conceptual framework and case analysis provided a strong foundation for the potential of blockchain use for AI governance. However, it is important to highlight several key considerations for the effective use and deployment of blockchain as a governance infrastructure for AI systems.

6.1 Blockchain data integrity and the oracle problem

The effectiveness of blockchain-based governance depends on the reliability of data entering the system. If inaccurate, biased, or manipulated data are recorded, blockchain may reinforce rather than mitigate governance risks. These risks are directly associated with the oracle problem, which results when data is inserted into the blockchain through

an oracle (e.g., human, IoT, etc.) rather than being created within the network itself as in the case of cryptocurrencies [75]. The oracle problem is one of the key limitations for blockchain use due to the inability of blockchain systems to verify and validate external real-world data [76]. Therefore, it is crucial for organizations to understand how data is fed into, used in and extracted from their AI systems and how it will be captured on-chain or off-chain and be integrated with the blockchain governance layer. Complementary mechanisms such as trusted execution environment and data pipelines, multi-source validation, and use of trusted third-party and human oversight can help with ensuring data integrity and reducing “the garbage in garbage out” risk of blockchain use. Some of these approaches are adopted by Prove AI and EQTY.

6.2 On-chain versus off-chain data storage and related design choices

As AI systems usually work with large datasets, blockchain data storage methods are another key consideration when it comes to the technology use for AI governance. While on-chain data storage offers data immutability, security and integrity, it comes with higher transaction costs and scalability challenges. Similarly, off-chain data storage reduces storage costs but risks data integrity and security, particularly in decentralized environments [77]. Therefore, it is important to consider the requirements and limitations of each data storage method when using blockchain. Hybrid approaches can help balancing the requirements for blockchain-governance layer in AI systems, in terms of system’s security, cost, performance and scalability. For instance, not all AI-related data can be stored on-chain but only governance-relevant metadata (e.g., model versions, audit logs, verification records) are recorded on-chain, while large datasets and computational processes remain off-chain with appropriate off-chain governance mechanisms. These design choices alongside decisions related to the frequency of blockchain data entry, data format and verification mechanism are context-dependent and require adaptation to specific organizational and regulatory settings. Therefore, they can vary per application domain.

6.3 Blockchain scalability and performance constraints

Blockchain scalability has been an outstanding challenge for the technology adoption and development [78]. Although this area has witnessed significant developments over the past years [79, 80], it remains a relevant factor to consider when using blockchain for large scale and evolving AI applications. Recording high-frequency AI lifecycle events

may introduce performance bottlenecks in the blockchain-governance layer, particularly in large-scale or real-time AI applications. This could lead to limitations in blockchain transaction throughput, latency, and storage capacity. Therefore, it is important to carefully evaluate and test blockchain system design, performance and data recording requirements when used in the governance of AI systems.

6.4 Governance of dynamic and evolving AI systems

AI systems are continuously updated, retrained, and/or repurposed, creating significant challenges for their governance. Blockchain technology may support versioning and traceability of AI model evolution, including changes in its input data, design parameters and outputs throughout its lifecycle [59]. However, considering the immutability of the technology, developed blockchain-based governance frameworks must account for continuous change and balance the use of on-chain versus off-chain data storage methods to ensure that lifecycle updates are meaningfully captured and validated. Furthermore, it is important to build resilient blockchain governance infrastructures that meet the evolving nature of AI systems, identifying and addressing blockchain possible vulnerabilities and system failure risks to ensure the sustainable and scalable use of the technology for AI governance.

6.5 Transparency versus confidentiality trade-offs

Blockchain promotes transparency, traceability and auditability, but sometimes this may conflict with the protection of proprietary data. Although the technology has been used to support intellectual property processes [12]. It is important to consider how the use of blockchain for AI governance will deal with propriety models, sensitive data, and intellectual property. The use of permissioned blockchain networks, having selective disclosure, and cryptographic verification can help balance the competing requirements of transparency and confidentiality [81]. Furthermore, the integration of blockchain and trusted execution environments can help in addressing the confidentiality requirements of sensitive AI models and applications. However, these approaches might add more complexity to the use of the technology, requiring comprehensive empirical testing.

6.6 Decentralization and governance control

The core capabilities of blockchain technology can play a critical role in enabling decentralized network governance [82]. However, to achieve its full potential, technical and managerial decisions need to be made to support decentralized network design and control. Decisions related to the

selection of consensus mechanism, access rights and use of smart contracts can influence the degree of decentralization in the governance system. In many cases, control may remain concentrated among a limited set of actors, particularly in enterprise environments [83]. This highlights that decentralization is not inherent but must be intentionally designed.

6.7 Economic and operational considerations

The inclusion of blockchain governance layer in AI systems will lead to infrastructural, operational, and financial changes that require comprehensive evaluation. Organizations must conduct cost–benefit analysis for the use of the technology to evaluate required infrastructure costs, coordination efforts, operational requirements and complexity, and the need for stakeholder alignment across the AI system lifecycle.

6.8 Usability and adoption challenges

Several users with variable levels of technical expertise interact with AI systems throughout their lifecycle including developers, auditors, regulators, and end-users. For blockchain to be effectively used as a governance layer for AI systems, it should be integrated seamlessly within the AI system and accessible to the different users. Complex architectures and technical barriers may hinder adoption, emphasizing the need for user-friendly interfaces and integration with existing workflows.

7 Discussion

The evolving nature of AI systems has stressed the need for robust yet adaptable governance mechanisms as AI systems are continuously retrained, updated, and integrated into evolving organizational and societal contexts. This has created tremendous challenges for governments aiming to develop appropriate regulatory frameworks that promote innovation and minimize adverse risks [11, 51]. Questions related to how to address AI accountability issues [21, 22], whether AI governance should be centralized or decentralized [84], and how organizations can govern AI systems throughout their entire lifecycle [10] remain only partially addressed.

Efforts to address AI governance challenges are fragmented. Existing knowledge in data governance [10] and information technology (IT) governance can support AI governance since most AI systems process large amounts of data to identify predictions, generate insights, and support decision-making, and this data is usually collected through

and shared with various IT systems [54]. AI systems also have to adhere to various regulations including general data protection regulations [85] and the health insurance portability and accountability act (HIPPA) [86], stressing the need for a better understanding of how existing regulations and governance mechanisms influence the development of AI systems and vice versa.

This paper adopts a normative and conceptual governance-oriented approach that positions blockchain not merely as a technical tool, but as a potential governance infrastructure capable of embedding transparency, accountability, traceability, and coordination mechanisms throughout the AI lifecycle. The proposed framework offers key insights on how blockchain can be relevant for AI governance at each stage of the AI system lifecycle, providing an initial guidance on the type of governance-related data that can be stored on the blockchain. The use of blockchain governance infrastructure provides a practical solution that can complement and support existing AI governance efforts.

Blockchain has proven to help address data governance issues in various contexts [15, 16] and several organizations are capitalizing on the properties of blockchain technology to create governance layer in AI systems in real-world setting [67, 69, 71]. Blockchain can help tackle the blackbox challenge and the opacity of AI models by improving the transparency, traceability and verifiability of AI systems data, models and outputs [58, 87]. It may also complement alternative AI governance approaches like federated learning, trusted execution environments and traditional audit logging systems, providing greater potential for the development of broader socio-technical AI governance ecosystems where different technologies complement each other depending on the context and application requirements.

The ability of blockchain to create a traceable immutable and verifiable digital audit trail is the cornerstone of its potential for AI governance, ensuring compliance with regulations, organizational policies, and ethical and societal norms and values. Blockchain has the potential to help tackle issues related to accountability in the use and development of AI systems, minimise adverse risks, protect human rights, and ensure responsible and authorized use of AI systems and related data [15]. The technology can support governance at the intraorganizational and interorganizational levels where an individual organization or a network of organizations can use the technology to track, monitor and promote the transparent and responsible development and use of AI systems in various application areas. The analysis of the three blockchain-enabled AI governance platforms (Prove AI, EQTY Lab, and Humans.ai) demonstrates that blockchain governance is already emerging in practice through different governance configurations, ranging from AI auditability and lifecycle verification to decentralized AI orchestration

and governance automation. The comparative analysis also highlights important limitations related to blockchain-AI scalability and interoperability, infrastructure dependency and the evolving nature of governance mechanisms in emerging blockchain-AI ecosystems.

Importantly, blockchain does not replace existing governance efforts but offers a mechanism for operationalizing them. For instance, decentralized SSI systems and verifiable credentials can support authenticated contribution, consent management, and tamper-proof certification of datasets and models, helping address long-standing issues of dataset lineage and model provenance [65]. This is particularly relevant for addressing the limitations of federated learning techniques. Blockchain-enabled decentralized autonomous organizations (DAOs) [88], can further support the development of a global AI governance infrastructure that enables decentralized decision-making and AI systems verification and validation across jurisdictions. These systems are particularly relevant in cross-border contexts where AI is developed and deployed across diverse institutional, legal, and cultural environments. Initiatives such as the Ministry of Foreign Affairs of the People's Republic of China (2023) and the United Nation Global Dialogue on AI Governance [43], may facilitate the development of global AI governance policies and principles capable of informing the development of blockchain-enabled AI governance infrastructures.

Besides investigating the potential of blockchain for AI governance, this study also identifies several key considerations influencing the feasibility and effectiveness of blockchain-enabled AI governance systems. First, the oracle problem remains a critical challenge because blockchain can only guarantee the integrity of the information recorded on-chain, not the validity of external data sources feeding the system. Therefore, organizations must develop appropriate mechanisms for improving data reliability before it is recorded on-chain and mitigate data integrity risks off-chain [89]. Second, practical blockchain governance architectures will likely require hybrid on-chain/off-chain models, where governance-relevant metadata are stored on-chain while computationally intensive processes and large datasets remain off-chain to address storage and performance limitations. Third, blockchain scalability remains an important consideration, particularly for large-scale and continuously evolving AI systems where recording high-frequency lifecycle events may introduce latency, storage overhead, and operational costs. Fourth, the potential use of multiple blockchain platforms for AI governance emphasize the integration and interoperability challenges between the different blockchain platforms and with other information systems [90]. Blockchain technology integration with AI and IoTs and industry 4.0 can also bring great potential but yet require further investigation [91]. The ongoing

developments of blockchain cross-chain interoperability solutions [92] can help when using blockchain for AI governance. Some blockchain architectures supports improved interoperability such as the Hedra blockchain network used in the case of EQTY Lab [68].

Beside the technical challenges, it is important to highlight key regulatory, contextual, societal and ethical considerations influencing blockchain use for AI governance. First, blockchain is an emerging technology with evolving regulatory frameworks and standards governing its use in different application areas and geographies [93]. Therefore, it is important to consider the implications of blockchain use and compliance requirements when integrated with AI systems. Organizations will need to make sure that any blockchain-based governance system follows relevant regulations as they evolve. Emerging regulatory frameworks such as the EU AI Act [38] and Regulation EU 2022/858 [94], on AI and distributed ledger technologies are increasingly offering clearer compliance pathways, which organizations can adopt as reference architectures to align blockchain-based governance practices with regulatory expectations. Having an adaptable yet secure design of the blockchain governance platforms where the different rules can be built into the blockchain itself can ease the adoption of the technology. Second, deploying blockchain-based governance infrastructures requires collective agreement among the various actors interacting with the AI system throughout its lifecycle. These actors may operate across different jurisdictions, regulatory systems, and institutional contexts, while also upholding different ethical, political, and cultural values. Differences in incentives, governance priorities, access rights, and technical capabilities can create imbalances in power structures and inclusivity, potentially limiting the adoption and legitimacy of blockchain-enabled governance systems. To address these challenges, blockchain-based governance systems must be designed to uphold human dignity, autonomy and inclusivity and reflect universal ethical values ensuring the transparency, inclusivity, accountability and responsible use of AI. As recent work highlights, embedding values is not only a technical design task, but a continuous process that requires shared norms and collective agreement among stakeholders throughout the AI system lifecycle [95]. In this sense, the success of blockchain-enabled AI governance will not only depend on the technical effectiveness of the technology but ultimately on whether the social, cultural, and institutional environments in which these systems are deployed actively support and reinforce shared governance values.

8 Conclusion and Future Research

This paper adopts a normative approach to examine the potential of blockchain technology in supporting AI governance. It critically examines how blockchain can address key risks in AI systems development and use while enabling the creation of practical, enforceable governance mechanisms throughout the AI system lifecycle. By providing a secure, traceable, and immutable audit trail, blockchain can enhance the transparency, accountability, and responsibility in AI systems without compromising innovation. Building on an analysis of AI system risks, existing governance initiatives, relevant governance and socio-technical literature and practical case studies, the study proposes a conceptual framework positioning blockchain as a governance infrastructure for AI systems. This framework provides initial guidance on the types of data that can be stored on the blockchain to enhance AI systems' governance, transparency, accountability, and compliance with relevant laws, regulations, organizational policies, and ethical standards.

The study also emphasizes key technical, organizational, societal, and ethical considerations that influence the effective use of blockchain for AI governance, including blockchain data integrity, scalability, regulation maturity, the need for coordination among network participants, and the need to uphold values such as autonomy, inclusivity, and human dignity.

While the study provides practical guidance and a governance-oriented framework, it is conceptual in nature and does not provide empirical validation or technical implementation. Future research is needed to validate the framework through real-world case studies, pilot implementations, and technical experimentation across diverse AI applications and organizational contexts. Specifically, future technical research should: (1) conduct comparative analysis of blockchain performance with alternative governance technologies, (2) evaluate blockchain's vulnerabilities and system risks, (3) examine blockchain design choices related to data capture and validation mechanism, the use of on-chain/off-chain data storage and blockchain accessibility for different users, (4) investigate data integrity issues and the oracle problem when integrating blockchain with the AI system, and (5) examine the scalability and interoperability of blockchain-AI systems considering continuous advancement in both technologies. Future studies should also investigate the economic, regulatory, and usability dimensions of blockchain-enabled AI governance systems considering related implementation and operational costs for blockchain use. This study lays the groundwork for operationalizing responsible AI governance, aiming to bridge the principles-to-practice gap. It provides valuable insights for policymakers, researchers, and practitioners

seeking to promote ethical, transparent, and accountable AI systems in diverse organizational and societal settings.

Author contribution W.A.H.A wrote the entire manuscript.

Data availability No datasets were generated or analysed during the current study.

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